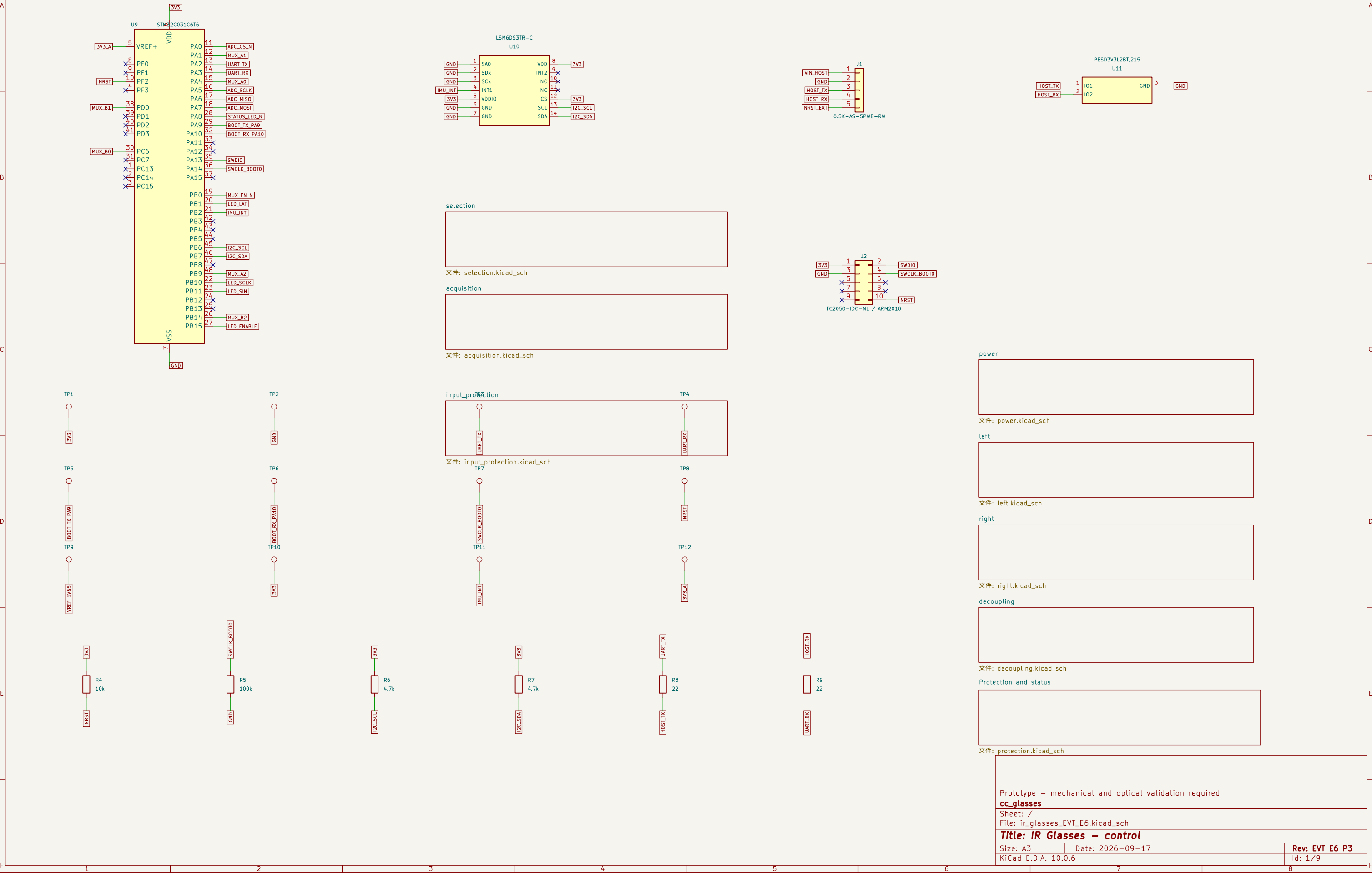
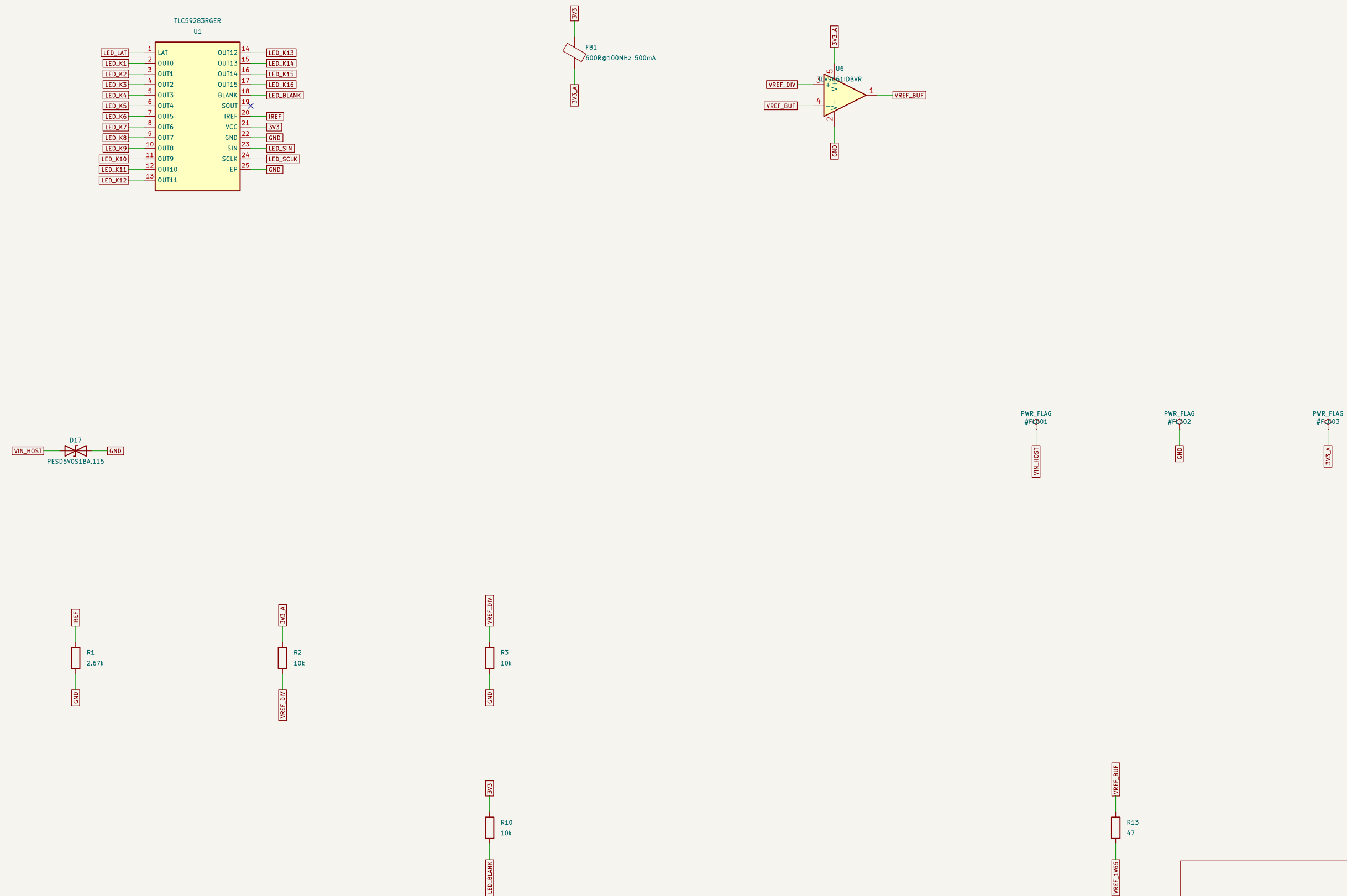


IR GLASSES EVT E6 – CONTROL



20 mA/ch; one LED at a time. BLANK pulled high during reset.

TIA: PD cathode at virtual ground, anode at GND; output rises with light.



Prototype – mechanical and optical validation required

cc_glasses

Sheet: /power/

File: power.kicad_sch

Title: IR Glasses – power

Size: A3

Date: 2026-09-17

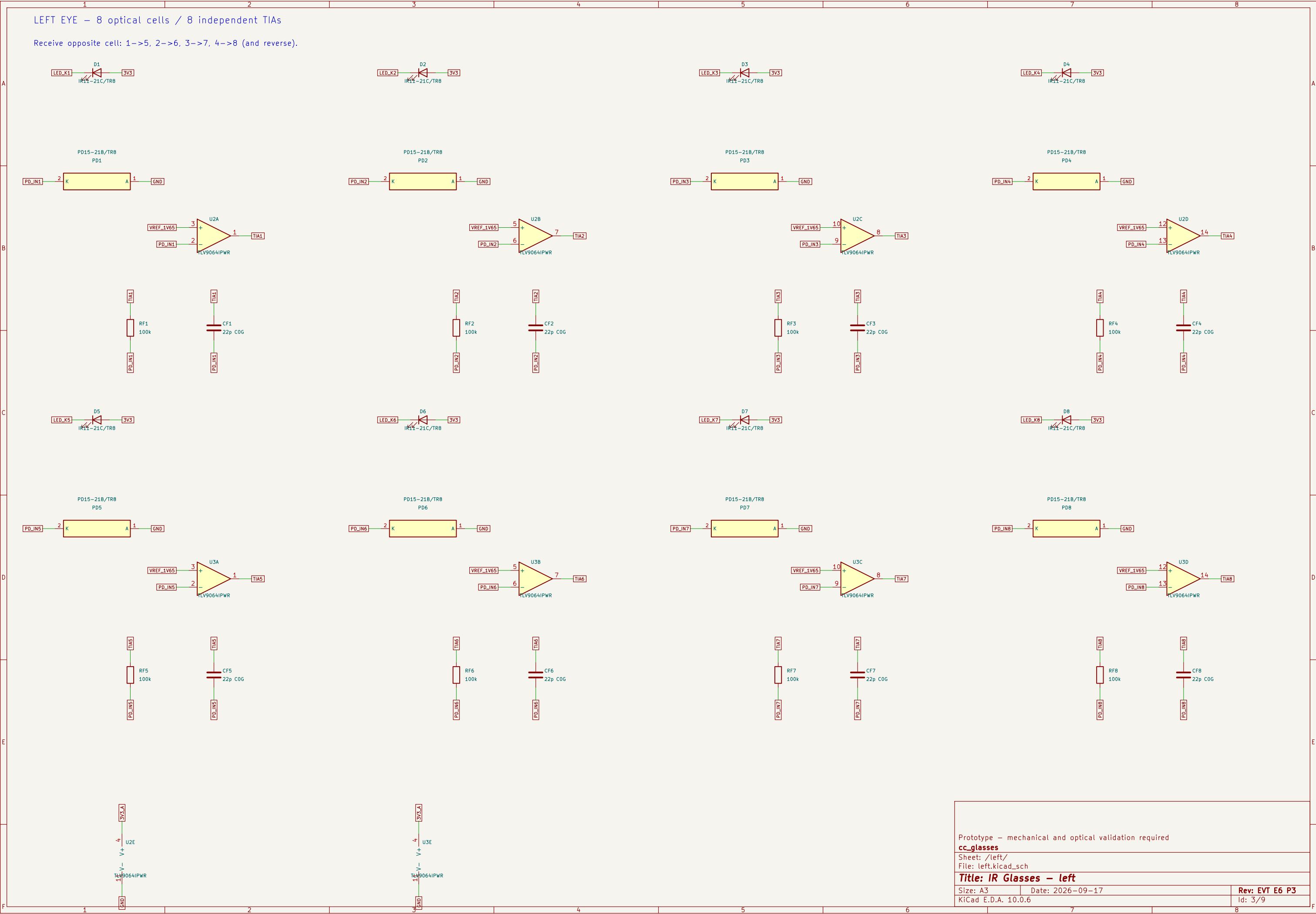
Rev: EVT E6 P3

KiCad E.D.A. 10.0.6

Id: 2/9

LEFT EYE – 8 optical cells / 8 independent TIAs

Receive opposite cell: 1->5, 2->6, 3->7, 4->8 (and reverse).



Prototype – mechanical and optical validation required

cc_glasses

Sheet: /left/

File: left.kicad_sch

Title: IR Glasses – left

Size: A3

Date: 2026-09-17

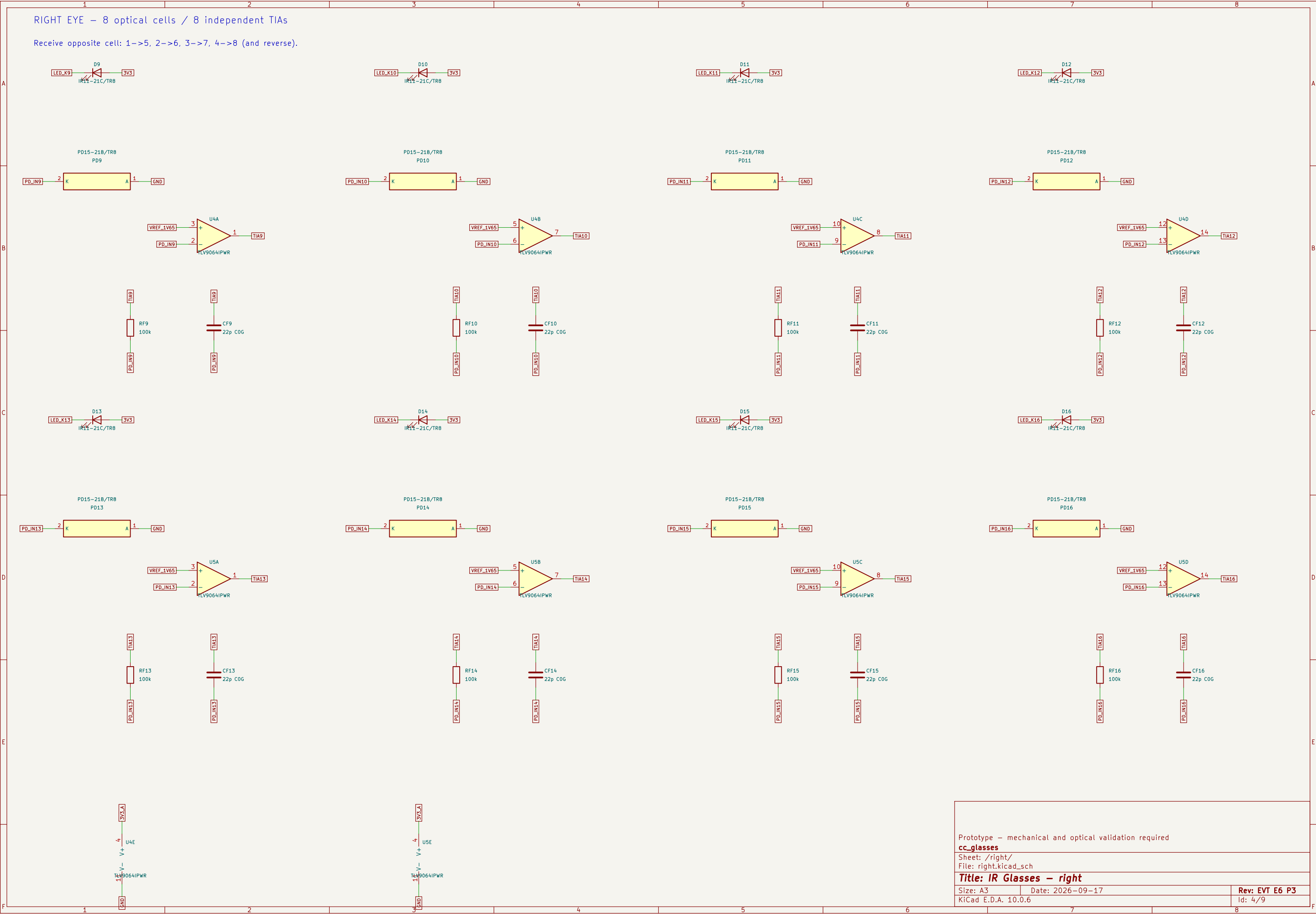
Rev: EVT E6 P3

KiCad E.D.A. 10.0.6

Id: 3/9

RIGHT EYE – 8 optical cells / 8 independent TIAs

Receive opposite cell: 1->5, 2->6, 3->7, 4->8 (and reverse).



Prototype – mechanical and optical validation required

cc_glasses

Sheet: /right/

File: right.kicad_sch

Title: IR Glasses – right

Size: A3

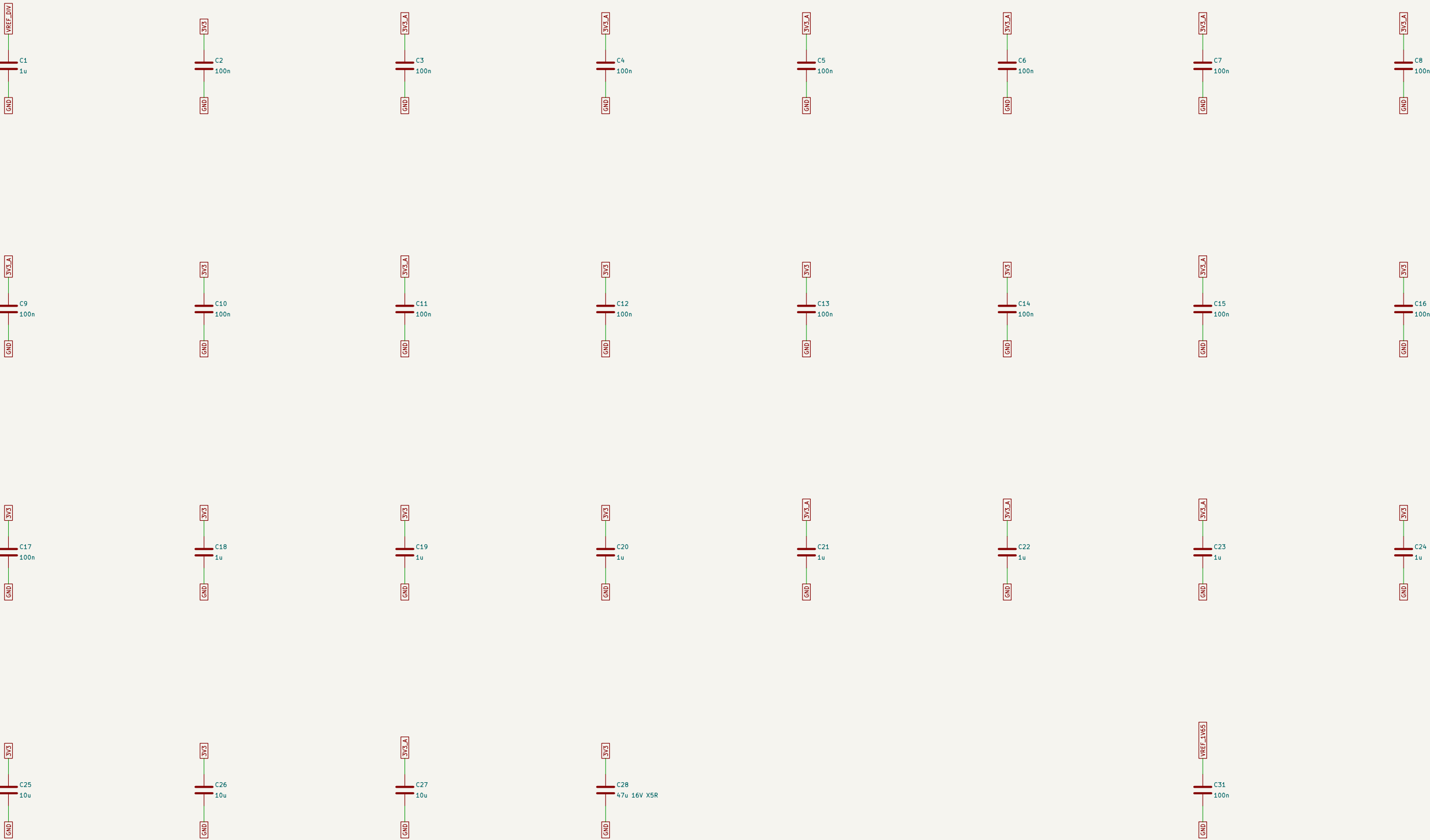
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Rev: EVT E6 P3

KiCad E.D.A. 10.0.6

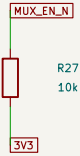
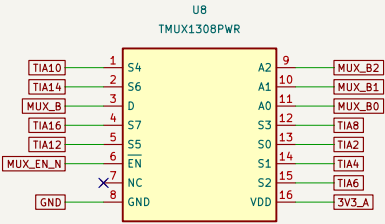
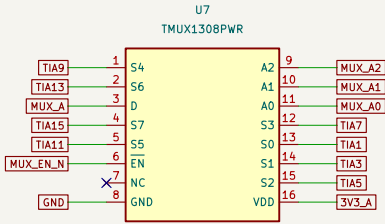
Id: 4/9

IR GLASSES EVT E6 – DECOUPLING





IR GLASSES EVT E6 – ADJACENT ODD / EVEN PD SELECTION



U7 selects odd PDs; U8 selects even PDs. LED_ID selects its two adjacent PDs using the fixed table.

EN_N is active low; R27 pulls high to disable both paths during reset.

Switch only with LED_ENABLE low. MUX address = (PD_ID – 1) / 2, integer division.

IR GLASSES EVT E6 – DUAL SIMULTANEOUS 16-BIT ACQUISITION

Each path: MUX -> unity buffer -> 10k/20k divider -> unity buffer -> 100R / 330p -> ADC.

AD7386: CH=0, SEQ=0, internal 2.5V reference, 1-wire output, 16-bit, no oversampling.

AINA0 and AINB0 sample together on ADC_CS_N falling edge. AINx1 grounded.

Analog span scaled 2/3. Calibrate path offset/gain; 16-bit codes do not guarantee 16-bit ENOB.

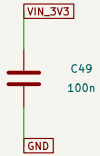
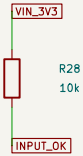
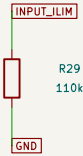
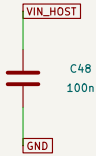
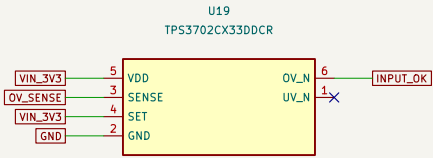
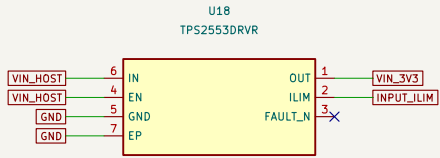
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Size: A3	Date: 2026-09-17	Rev: EVT E6 P3
KiCad E.D.A. 10.0.6		Id: 8/9

Each path: MUX -> unity buffer -> 10k/20k divider -> unity buffer -> 100R / 330p -> ADC.

AD7386: CH=0, SEQ=0, internal 2.5V reference, 1-wire output, 16-bit, no oversampling.

AINA0 and AINB0 sample together on ADC_CS_N falling edge. AINx1 grounded.

INPUT PROTECTION – 3.3 V NOMINAL / CORRECT POLARITY



J1 -> U18 (110k current limit) -> VIN_3V3 -> U14 -> 3V3.

U19 TPS3702CX33: SET=VIN; R18=200R / R19=10k; OV_N -> U14 EN.

U14 OVLO=GND. EN low disables; recovery follows the C38=22nF startup path.

Nominal OV trip 3.501V / release 3.481V. Input: 3.3V +/-3%, correct polarity.

No guaranteed fast surge clamp: comparator and EN response have delays.

Bench validation required: startup, hot-plug, recovery, short circuit and temperature.